

The Telegraph Model of Gene Expression: A Triangular Cross-Validation of Stochastic Simulation

Zakhar Guzii

Kyiv School of Economics, Department of Mathematics
zakharguzii@gmail.com

June 2026

Abstract

Gene expression—the process by which a gene’s DNA is read out to make messenger RNA (mRNA), the molecule that carries its instructions—is a stochastic process: even genetically identical cells in the same environment produce very different numbers of mRNA molecules. The telegraph model explains the main source of this variability. It treats a gene as a two-state (ON/OFF) switch that transcribes mRNA only while active. We study the telegraph model in three independent ways and check that all three agree: we (i) generate stochastic trajectories with the Gillespie stochastic simulation algorithm (SSA); (ii) derive and solve a closed system of ordinary differential equations (ODEs) for the first and second moments; and (iii) draw independent samples directly from the analytic steady-state distribution, using the classic Beta–Poisson law together with an explicit Bernoulli gene state. The main contribution is the cross-validation that links the three methods. The code is available at <https://github.com/zakhar-guzii/telegraph-model-gene-expression>.

1 Introduction

A surprising fact of molecular biology is that gene expression is variable. Two cells with the same DNA, grown together in the same environment, can have different numbers of copies of the same mRNA or protein [6, 7]. This difference between cells is called phenotypic noise. It happens naturally because of random molecular interactions. Experimentally this variability separates into two parts [6]: *extrinsic* noise, caused by differences between cells—their size, cell-cycle stage, or the pool of available polymerases—and *intrinsic* noise, the randomness of the transcription and degradation events themselves, which persists even between two identical copies of a gene inside the same cell.

Inside a cell, the key players—a single copy of the gene, a handful of transcription factors (regulatory proteins that bind DNA and control gene activity), and a few dozen mRNA molecules—are present in such small numbers that the discreteness and randomness of individual chemical events cannot be averaged away. The telegraph model studied here is the simplest framework that describes this intrinsic component. It idealises the promoter (the DNA region that controls when a gene is switched on) as a two-state device—like a telegraph key that randomly switches between an active (ON) and an inactive (OFF) state [8]. While ON, it produces mRNA molecules through transcription in random bursts; each molecule is then degraded at a random time. As a result, the mRNA count is a discrete-valued stochastic process.

Despite its simplicity, the telegraph model reproduces the bursty, *over-dispersed* mRNA distributions seen in single-molecule experiments, which count individual mRNAs inside single cells [5], and it can be analysed exactly [4]. A convenient way to measure this spread is the *Fano factor* $F = \sigma^2/\mu$ [9], the ratio of the variance to the mean of the mRNA count. Pure Poisson production and decay, in which every event is independent, has variance equal to mean, giving the *Poisson baseline* $F = 1$. “Over-dispersed” means $F > 1$: the random switching of the gene adds extra variability on top of that baseline.

We treat the telegraph model as a case study in how a stochastic process can be described from several mathematically distinct viewpoints. We implement three approaches that look unrelated:

1. a stochastic simulation (the SSA), which answers “what does a single cell actually do?”;
2. a system of moment ODEs, which answers “what does the average cell do, and how spread out is the population?”;
3. a steady-state sampler, which answers “what does the population look like after a very long time?”.

Each approach is correct by itself. But only by comparing them can we be sure that all three are built correctly and that our understanding of the model is right.

2 Model Formulation

The state of the system at time t is the pair

$$(G(t), R(t)), \quad G(t) \in \{0, 1\}, \quad R(t) \in \mathbb{N}, \quad (1)$$

where the binary variable G records whether the gene is OFF ($G = 0$) or ON ($G = 1$), and the integer R counts the number of mRNA molecules present. Four basic reactions define the dynamics of the system. Each reaction has a *propensity* a_j , the rate at which it fires at any given moment:

Reaction	Description	Effect	Propensity
R_1	Gene activation	$G : 0 \rightarrow 1$	$a_1 = k_{\text{on}} (1 - G)$
R_2	Gene inactivation	$G : 1 \rightarrow 0$	$a_2 = k_{\text{off}} G$
R_3	RNA synthesis	$R \rightarrow R + 1$	$a_3 = k_{\text{syn}} G$
R_4	RNA degradation	$R \rightarrow R - 1$	$a_4 = k_{\text{deg}} R$

All four rate constants are positive. The switching rates $k_{\text{on}}, k_{\text{off}}$ set how quickly the promoter turns on and off, and their ratio fixes the long-run fraction of time the gene is active; the factor G in a_3 shuts transcription off entirely when the gene is OFF; and k_{deg} is a per-molecule decay rate, so the total degradation propensity $k_{\text{deg}} R$ grows with the amount of mRNA present. Together the rates $(k_{\text{on}}, k_{\text{off}}, k_{\text{syn}}, k_{\text{deg}})$ and an initial condition (g_0, r_0) at time t_0 fully determine the law of the process.

Burst size. One derived quantity is used repeatedly below and follows in a line. Once the gene is ON, the only reaction that can switch it off fires at the constant rate k_{off} , so the duration of an active period is exponential with mean $1/k_{\text{off}}$; during that period transcription fires at rate k_{syn} . The mean number of transcripts made per active period—the *burst size* B —is therefore

$$B = k_{\text{syn}} \mathbb{E}[\text{ON duration}] = \frac{k_{\text{syn}}}{k_{\text{off}}}. \quad (2)$$

Bursts of this kind are observed directly in single cells [10, 5], and the exact consequences of promoter switching for the mRNA distribution, including the limits recovered in Section 4.1, are worked out in [11].

Because each waiting time is exponential and depends only on the current state, $(G(t), R(t))$ is a continuous-time Markov chain (CTMC). Its distribution $P_t(g, r) = \Pr\{G(t) = g, R(t) = r\}$ evolves according to the chemical master equation [13, 14, 15, 16]

$$\begin{aligned} \frac{d}{dt}P_t(g, r) = & k_{\text{on}}[\delta_{g,1}P_t(0, r) - \delta_{g,0}P_t(g, r)] \\ & + k_{\text{off}}[\delta_{g,0}P_t(1, r) - \delta_{g,1}P_t(g, r)] \\ & + k_{\text{syn}}\delta_{g,1}[P_t(1, r-1) - P_t(1, r)] \\ & + k_{\text{deg}}[(r+1)P_t(g, r+1) - rP_t(g, r)], \end{aligned} \quad (3)$$

with the convention $P_t(g, -1) = 0$. Equation (3) underlies all three approaches below: the SSA samples trajectories whose distribution solves it, the moment ODEs are derived from it exactly, and the steady-state sampler draws from its $t \rightarrow \infty$ fixed point.

3 Methods

3.1 Stochastic Simulation (Gillespie SSA)

The *stochastic simulation algorithm* (SSA) of Gillespie [1, 2, 3] generates statistically exact sample paths of a CTMC, making no approximation to the master equation (3). Its key idea is that, given the current state, two questions separate: when does the next reaction occur, and which reaction is it? With $a_0 = \sum_{j=1}^4 a_j$ the *total propensity*, the waiting time is $\tau \sim \text{Exp}(a_0)$ and reaction j fires with probability a_j/a_0 .

Our implementation draws these by inversion, using two independent uniforms $r_1, r_2 \sim \text{U}(0, 1)$ per step:

$$\tau = \frac{1}{a_0} \ln \frac{1}{r_1}, \quad j = \min \left\{ k : \sum_{i=1}^k a_i > r_2 a_0 \right\}. \quad (4)$$

The time is then advanced by τ , the state updated by reaction j , the reaction propensities recomputed, and the loop repeats for a fixed number of events n_{sim} . If the total propensity reaches $a_0 = 0$, no reaction can fire, so the state is absorbing and is held fixed afterwards.

A single trajectory is the history of one cell. To obtain population statistics, we run N_{rep} independent trajectories with identical parameters and average over them. Each trajectory advances by its own random time increments, so the same event index occurs at different times in different trajectories. We therefore resample every trajectory onto a shared uniform time grid by a zero-order hold before averaging; the state is piecewise constant between events, so this introduces no error. The empirical moments are then ensemble averages at each grid time, for example

$$\widehat{\mu_R}(t) = \frac{1}{N_{\text{rep}}} \sum_{i=1}^{N_{\text{rep}}} R_i(t), \quad \widehat{\mu_G}(t) = \frac{1}{N_{\text{rep}}} \sum_{i=1}^{N_{\text{rep}}} G_i(t), \quad (5)$$

with sample variances and the sample gene–RNA covariance defined analogously.

Simulation horizon. The loop counter n_{sim} bounds the number of *events*, not the elapsed time, so an ensemble has no common end time. We therefore take the shared grid no further than the earliest-finishing trajectory, $t_{\text{end}} = \min_i t_i^{\text{last}}$: extending past that point would freeze the exhausted

trajectories at their final state and bias the ensemble toward whatever those particular cells happened to be doing when they ran out of events. Each figure therefore ends at its own, seed-dependent time, in every case a large multiple of the relaxation time.

Figure 1 illustrates a typical sample path generated by the algorithm at the reference parameters $k_{\text{on}} = k_{\text{off}} = 0.5$, $k_{\text{syn}} = 20$, $k_{\text{deg}} = 1$. Averaging $N_{\text{rep}} = 1000$ such trajectories according to (5) gives the ensemble moments that Section 4.2 compares with the moment ODEs of Section 3.2.

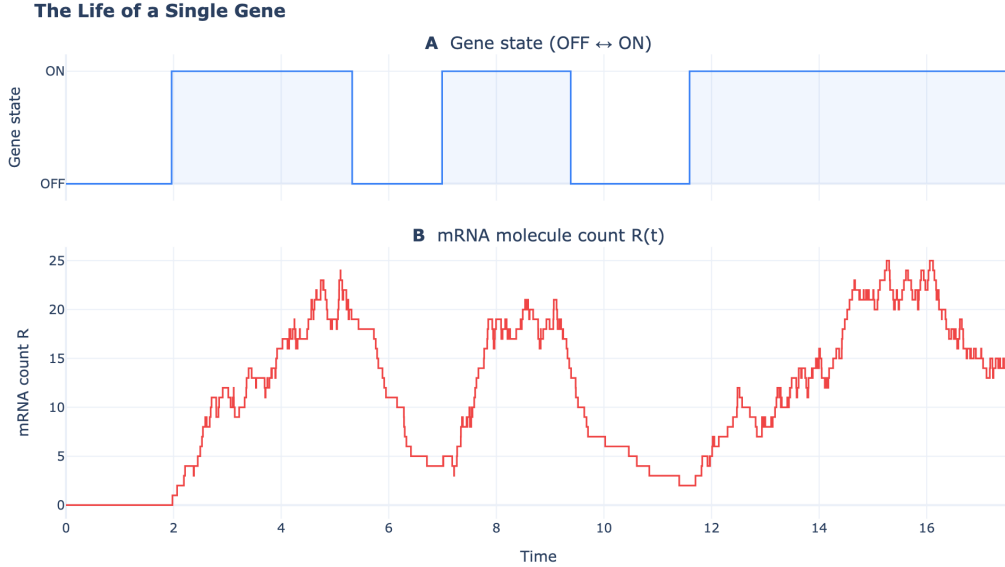


Figure 1: One stochastic run, i.e. a single sample path (“the life of a single gene”), at $k_{\text{on}} = k_{\text{off}} = 0.5$, $k_{\text{syn}} = 20$, $k_{\text{deg}} = 1$. Top: the gene state switches between OFF and ON at random times. Bottom: the mRNA count rises in bursts during ON periods and decays between them. Averaging many such trajectories gives the sample moments used below.

Random number generation. All experiments draw from the Mersenne Twister [20] in its MT19937 form, as supplied by `numpy.random` [17], with a seed fixed for each experiment.

3.2 Moment ODEs

The SSA describes individual cells. Often we instead want deterministic equations for population-level statistics—the *moments*. A moment is an expectation: the mean $\mu_R = \mathbb{E}[R]$ is the average mRNA level, the variance $\sigma_R^2 = \mathbb{E}[R^2] - \mathbb{E}[R]^2$ measures the noise, and the covariance $C_{RG} = \mathbb{E}[RG] - \mathbb{E}[R]\mathbb{E}[G]$ measures how gene activity and mRNA level vary together.

Applying the standard moment rule for a chemical master equation—the expected rate of change of any observable is the sum over reactions of each propensity times the change it makes—to

$f = G, R, R^2, GR$ in turn yields four coupled equations for the telegraph model [16, 15]:

$$\frac{d\mu_G}{dt} = k_{\text{on}} (1 - \mu_G) - k_{\text{off}} \mu_G, \quad (6)$$

$$\frac{d\mu_R}{dt} = k_{\text{syn}} \mu_G - k_{\text{deg}} \mu_R, \quad (7)$$

$$\frac{d\sigma_R^2}{dt} = 2k_{\text{syn}} C_{RG} - 2k_{\text{deg}} \sigma_R^2 + k_{\text{syn}} \mu_G + k_{\text{deg}} \mu_R, \quad (8)$$

$$\frac{dC_{RG}}{dt} = k_{\text{syn}} \sigma_G^2 - (k_{\text{on}} + k_{\text{off}} + k_{\text{deg}}) C_{RG}. \quad (9)$$

Each has a direct reading. Activation pushes mean activity toward 1 and inactivation pulls it toward 0; mRNA is made in proportion to the active fraction μ_G and removed in proportion to the amount present; the mRNA variance is driven up by the coupling to gene activity (the C_{RG} term) and by the randomness of production and decay, and down by degradation; and the covariance is created by transcription passing the gene's own fluctuations σ_G^2 on to the mRNA, then damped at the combined rate of switching and degradation.

Exact closure. In a general nonlinear reaction system the equation for each moment depends on higher moments, so the system never closes and one has to add approximate closure schemes [12, 16, 15]. Here it closes *exactly*. The only possible nonlinearity is the variance of the gene state, but because G is binary,

$$\sigma_G^2 = \mathbb{E}[G^2] - \mu_G^2 = \mathbb{E}[G] - \mu_G^2 = \mu_G (1 - \mu_G), \quad (10)$$

which depends on the mean μ_G alone. Putting (10) into (9) turns the four equations (6)–(9) into a *closed, exact* linear system in $(\mu_G, \mu_R, \sigma_R^2, C_{RG})$, whose solutions are the true moments of the master equation (3). A four-variable numerical integration is therefore sufficient, with no closure approximation anywhere in the code.

Implementation. The four right-hand sides (6)–(9), with σ_G^2 substituted by (10), are coded directly and passed to the explicit Runge–Kutta solver `scipy.integrate.solve_ivp` [18] (method RK45, the embedded Dormand–Prince pair [19]) from the initial conditions $\mu_G(t_0) = g_0$, $\mu_R(t_0) = r_0$, $\sigma_R^2(t_0) = C_{RG}(t_0) = 0$, evaluated on a uniform grid of 1000 time points. Section 4.2 overlays them on the SSA ensemble at the same rates and the same initial condition.

Steady-state Fano factor. Setting the right-hand side of (9) to zero gives $C_{RG}^{ss} = k_{\text{syn}} \sigma_G^2 / (k_{\text{on}} + k_{\text{off}} + k_{\text{deg}})$ with $\sigma_G^2 = \mu_G^{ss}(1 - \mu_G^{ss})$ by (10); substituting that into (8) at stationarity, where $2k_{\text{deg}} \sigma_R^2 = 2k_{\text{syn}} C_{RG} + k_{\text{syn}} \mu_G + k_{\text{deg}} \mu_R$ and $k_{\text{syn}} \mu_G = k_{\text{deg}} \mu_R$ by (7), and dividing by μ_R^{ss} yields the Fano factor in closed form:

$$F^{ss} = \left. \frac{\sigma_R^2}{\mu_R} \right|_{ss} = 1 + \frac{k_{\text{syn}} k_{\text{off}}}{(k_{\text{on}} + k_{\text{off}})(k_{\text{on}} + k_{\text{off}} + k_{\text{deg}})}. \quad (11)$$

The additive 1 is the Poisson baseline: it is what survives when $k_{\text{off}} \rightarrow 0$ and the gene stops switching, so the mRNA count is a pure birth–death process. The second term is the excess contributed by promoter switching, and it vanishes in two distinct ways—as $k_{\text{off}} \rightarrow 0$ (the gene is always ON) and as $k_{\text{on}}, k_{\text{off}} \rightarrow \infty$ at fixed ratio, where the denominator grows quadratically while the numerator grows linearly, so a fast-flickering promoter averages out. Because the decay in that second limit is only $\mathcal{O}(1/k_{\text{on}})$, the approach to $F = 1$ is slow; Table 1 quantifies it.

3.3 Steady-State Sampler

As $t \rightarrow \infty$ the telegraph model forgets its initial condition and converges to a unique stationary distribution $P_\infty(g, r)$. Instead of simulating a long transient to reach it, we can sample from P_∞ directly. Peccoud and Ycart [4] showed that at steady state the mRNA count follows the classic *Beta-Poisson* law: it is Poisson, but with a mean that is itself random from cell to cell. Let the *mixing fraction* $p \in [0, 1]$ be the long-run share of time a cell's gene spends ON. Across a population p is Beta-distributed, and given p the mRNA count is Poisson with mean $\lambda = cp$, where $c = k_{\text{syn}}/k_{\text{deg}}$ is the *transcription efficiency coefficient*: the mean mRNA level a permanently active gene would reach. With the switching rates $a = k_{\text{on}}/k_{\text{deg}}$ and $b = k_{\text{off}}/k_{\text{deg}}$ (so k_{deg} fixes the unit of time), the stationary mRNA law is therefore

$$R \sim \text{Poisson}(cp), \quad p \sim \text{Beta}(a, b). \quad (12)$$

Averaging the Poisson kernel over the Beta density gives this stationary mRNA law in closed form:

$$P_\infty(r) = \frac{c^r e^{-c}}{r!} \frac{\Gamma(a+r) \Gamma(a+b)}{\Gamma(a) \Gamma(a+b+r)} {}_1F_1(b, a+b+r; c), \quad (13)$$

where ${}_1F_1$ is Kummer's confluent hypergeometric function. Equation (13) is the Peccoud–Ycart result [4]; applying Kummer's transformation ${}_1F_1(\alpha, \gamma, z) = e^z {}_1F_1(\gamma - \alpha, \gamma, -z)$ with $\alpha = b$, $\gamma = a + b + r$, $z = c$ absorbs the e^{-c} prefactor and returns the equivalent form ${}_1F_1(a+r, a+b+r; -c)$ in which the law is more often quoted.

We record (13) but do not evaluate it directly. Because the mixture (12) *is* this distribution, sampling the mixture already samples P_∞ exactly, with no discretisation of the state space. This costs just three vectorised draws per count, avoiding the need to evaluate ${}_1F_1$. The closed form states what the sampler targets; it is not the algorithm.

Our sampler implements exactly this law, and draws the gene state as an explicit Bernoulli(p) so we can also estimate the joint moments of (G, R) . For each of the N_{rep} samples it draws

$$p \sim \text{Beta}(a, b), \quad f(p) = \frac{\Gamma(a+b)}{\Gamma(a)\Gamma(b)} p^{a-1} (1-p)^{b-1}, \quad (14)$$

$$G \mid p \sim \text{Bernoulli}(p), \quad \Pr(G = 1 \mid p) = p, \quad \Pr(G = 0 \mid p) = 1 - p, \quad (15)$$

$$R \mid p \sim \text{Poisson}(\lambda), \quad \lambda = \frac{k_{\text{syn}}}{k_{\text{deg}}} p. \quad (16)$$

The key point is that the mRNA count R is conditioned on the mixing fraction p , *not* on the current gene state G . Given p , the gene state and the mRNA count are drawn independently, so a cell can have $G = 0$ and still $R > 0$ —as in a real cell, where mRNA made during the last active period outlives the switch to OFF.

Validity of the joint construction. Peccoud and Ycart [4] derived the *marginal* law of R , given by (13). However, the checks below require evaluating $\widehat{\sigma}_G^{ss}$ and $\widehat{C}_{RG}^{ss}$, which are properties of the joint distribution. To achieve this, we must explicitly couple the gene state G to the model. A natural formulation is to assume that $G \sim \text{Bernoulli}(p)$ uses the same mixing fraction p that drives the Poisson mean, and that G and R are conditionally independent given p . We can verify the validity of this joint construction by checking whether it recovers the exact covariance derived directly from the master equation.

Setting (9) to zero and substituting the exact closure (10) yields the steady-state covariance from the master equation:

$$C_{RG}^{ss} = \frac{k_{\text{syn}} \sigma_G^2}{k_{\text{on}} + k_{\text{off}} + k_{\text{deg}}} = \frac{k_{\text{syn}} \mu_G^{ss} (1 - \mu_G^{ss})}{k_{\text{on}} + k_{\text{off}} + k_{\text{deg}}}. \quad (17)$$

Alternatively, computing the covariance directly from the sampler's conditional independence gives $\mathbb{E}[RG] = \mathbb{E}[\mathbb{E}[R | p] \mathbb{E}[G | p]] = c \mathbb{E}[p^2]$. Since $\mathbb{E}[R] \mathbb{E}[G] = c \mathbb{E}[p]^2$, it follows that $C_{RG} = c \text{Var}(p)$. Using the Beta variance $\text{Var}(p) = ab / ((a+b)^2(a+b+1))$ and substituting $ab / (a+b)^2 = \mu_G^{ss} (1 - \mu_G^{ss})$, $a + b + 1 = (k_{\text{on}} + k_{\text{off}} + k_{\text{deg}}) / k_{\text{deg}}$, and $c = k_{\text{syn}} / k_{\text{deg}}$, this expands to:

$$C_{RG} = \frac{k_{\text{syn}}}{k_{\text{deg}}} \cdot \frac{\mu_G^{ss} (1 - \mu_G^{ss}) k_{\text{deg}}}{k_{\text{on}} + k_{\text{off}} + k_{\text{deg}}} = \frac{k_{\text{syn}} \mu_G^{ss} (1 - \mu_G^{ss})}{k_{\text{on}} + k_{\text{off}} + k_{\text{deg}}}, \quad (18)$$

which matches (17) term for term. This confirms that the joint sampling scheme correctly captures the joint moments of the master equation.

Furthermore, this formulation behaves intuitively at the limits of the parameter space. As the switching rates become very slow compared with degradation ($a, b \rightarrow 0$), the $\text{Beta}(a, b)$ density concentrates its mass at the endpoints $p = 0$ and $p = 1$. The mixture then separates into its building blocks: a single Bernoulli trial for the gene state and, when ON, an ordinary Poisson count for the mRNA. Specifically, at $k_{\text{off}} = 0$, the gene never switches off ($p \equiv 1$), and the stationary law collapses directly to a plain $\text{Poisson}(k_{\text{syn}}/k_{\text{deg}})$ distribution. We test this limiting behaviour numerically in Section 4.3.

The joint stationary law is the average of these conditional distributions over the mixing fraction,

$$P_{\infty}(g, r) = \int_0^1 \Pr(G = g | p) \Pr(R = r | p) f(p) dp, \quad (19)$$

which our sampler never evaluates directly: drawing N_{rep} triples (p, G, R) and averaging gives unbiased Monte Carlo estimates of the steady-state moments $\widehat{\mu}_G^{ss}, \widehat{\mu}_R^{ss}$, and so on. Two edge cases are handled explicitly. If $k_{\text{off}} = 0$ the gene never switches off, so p is fixed at 1 rather than drawn; if $k_{\text{deg}} = 0$ no steady state exists (mRNA grows without bound) and the input is rejected. The exact stationary means used as the reference below are

$$\mu_G^{ss} = \frac{k_{\text{on}}}{k_{\text{on}} + k_{\text{off}}}, \quad \mu_R^{ss} = \frac{k_{\text{syn}}}{k_{\text{deg}}} \mu_G^{ss}, \quad (20)$$

obtained by setting the time derivatives in (6)–(7) to zero.

3.4 The Validation Triangle

We state these as the three edges of a *validation triangle* and test each one numerically in the experiments that follow.

Edge 1 — SSA $\leftrightarrow$ Moment ODEs (finite time). The empirical averages (5) are sample means of independent, identically distributed trajectories. By the Law of Large Numbers, as $N_{\text{rep}} \rightarrow \infty$ they converge to the true expectations, which are exactly the solutions of the moment ODEs:

$$\lim_{N_{\text{rep}} \rightarrow \infty} \frac{1}{N_{\text{rep}}} \sum_{i=1}^{N_{\text{rep}}} R_i(t) = \mu_R(t), \quad (21)$$

and likewise for $\mu_G(t)$ and the covariance $C_{RG}(t)$. This links the stochastic and deterministic descriptions: with finitely many replicates, the SSA should follow the ODE curves up to a statistical error of order $1/\sqrt{N_{\text{rep}}}$.

Edge 2 — Moment ODEs $\leftrightarrow$ Steady-state sampler (long time). The stationary distribution is the $t \rightarrow \infty$ limit of the dynamics, so the moment ODEs must approach a fixed point,

$$\lim_{t \rightarrow \infty} \mu_G(t) = \mu_G^{ss}, \quad \lim_{t \rightarrow \infty} \mu_R(t) = \mu_R^{ss}, \quad \text{etc.}, \quad (22)$$

which coincides with (20). The independent steady-state sampler must estimate *these same* values,

$$\lim_{N_{\text{rep}} \rightarrow \infty} \frac{1}{N_{\text{rep}}} \sum_{i=1}^{N_{\text{rep}}} R_i^{ss} = \mu_R^{ss}, \quad (23)$$

and likewise for μ_G^{ss} . By the Central Limit Theorem, the sampler's error decays as $\sigma/\sqrt{N_{\text{rep}}}$, where σ is the standard deviation of the stationary distribution. Observing this $1/\sqrt{N_{\text{rep}}}$ rate confirms that the sampler is a valid Monte Carlo estimator of the steady-state moments.

Edge 3 — SSA $\leftrightarrow$ Steady-state sampler (closing the triangle). An SSA trajectory run long enough to reach stationarity samples the same distribution P_∞ as the direct sampler. Hence the long-time plateau of the SSA moments must agree with both the ODE fixed point and the sampler estimates. Because both endpoints of this edge emit samples, the claim can be sharpened to the equality of the full laws, $\hat{P}_\infty^{\text{SSA}} = \hat{P}_\infty^{\text{smp}}$, rather than of their first two moments alone; Section 4.3 tests it in total-variation distance.

4 Experiments and Results

The study is organised into three Jupyter notebooks, one for each part of the validation.

Unless stated otherwise, the dynamic experiments use the symmetric parameters $k_{\text{on}} = 0.5$, $k_{\text{off}} = 0.5$, $k_{\text{syn}} = 20.0$, $k_{\text{deg}} = 1.0$, with $N_{\text{rep}} = 1000$ trajectories and initial condition $(g_0, r_0) = (0, 0)$. These give an active fraction $\mu_G^{ss} = k_{\text{on}}/(k_{\text{on}} + k_{\text{off}}) = 0.5$ and steady-state mean mRNA level $\mu_R^{ss} = (k_{\text{syn}}/k_{\text{deg}}) \mu_G^{ss} = 10$, by (20). The five-moment sampler check in Section 4.3 uses $k_{\text{off}} = 1$ instead, which gives $\mu_R^{ss} = 6.667$.

4.1 Parameter Exploration (Notebook 1)

Aim. This part runs the SSA on its own to check that it reproduces the expected behaviour when the gene's parameters change. We vary one mechanism at a time so that each run isolates a single effect (Table 1). The final run deliberately alters numerical settings rather than biological mechanisms, verifying the stability of the simulation itself.

Result. Three mathematical insights emerge from these runs. First, as switching rates grow relative to k_{deg} (Figure 2), the gene state averages out. Consequently, σ_R^2 and C_{RG} fall, and the measured Fano factor F approaches the Poisson baseline $F = 1$ (Table 1). At finite rates, the limit is approached but never fully reached, which perfectly aligns with the exact limits derived in [11].

Second, with k_{on} fixed, the burst size (2) dictates the steady-state mean μ_R^{ss} . As seen in the expression runs, a weaker promoter generates fewer molecules but exhibits a larger relative spread (coefficient of variation, σ_R/μ_R), explaining why rare-molecule genes appear noisier.

Third, the steady state is independent of both initial conditions and numerical resolution. The matching outcomes from different initialisations confirm the chain is ergodic. Furthermore, increasing the ensemble size (N_{rep}) simply reduces sampling error without shifting the steady state. Here, $N_{\text{rep}} = 224$ reflects typical cell counts in single-molecule mRNA studies [5]. Notebook 1 plots the moment traces behind all six runs.

Experiment	Parameters	F^{ss}	$\hat{F}$ (SSA)	What it isolates
Slow switching	$k_{\text{on}} = k_{\text{off}} = 0.05, k_{\text{syn}} = 20$	10.09	9.94	noise at fixed mean
Fast switching	$k_{\text{on}} = k_{\text{off}} = 10, k_{\text{syn}} = 20$	1.48	1.48	noise at fixed mean
High expression	$k_{\text{on}} = 0.5, k_{\text{off}} = 4, k_{\text{syn}} = 56$	10.05	9.90	burst size ≈ 14
Low expression	$k_{\text{on}} = 0.5, k_{\text{off}} = 10, k_{\text{syn}} = 30$	3.48	3.49	burst size ≈ 3
Initial cond.	$(g_0, r_0) = (0, 0)$ vs. $(1, 15)$	9.33	9.41 / 9.27	forgetting the start
Numerical	$n_{\text{sim}} = 1000$ vs. $N_{\text{rep}} = 224, n_{\text{sim}} = 5000$	9.33	9.48 / 9.40	estimate quality

Table 1: The six parameter-exploration runs. Unless shown otherwise, $k_{\text{deg}} = 1$ and the base switching rates are $k_{\text{on}} = k_{\text{off}} = 0.1$; each run reaches the steady-state mean $\mu_R^{ss} = 10$ except the expression runs, which change it on purpose. F^{ss} is the closed-form steady-state Fano factor (11); $\hat{F} = \hat{\sigma}_R^2 / \hat{\mu}_R$ is measured from the corresponding SSA run, averaged over the last fifth of the horizon. Rows with two runs report both.

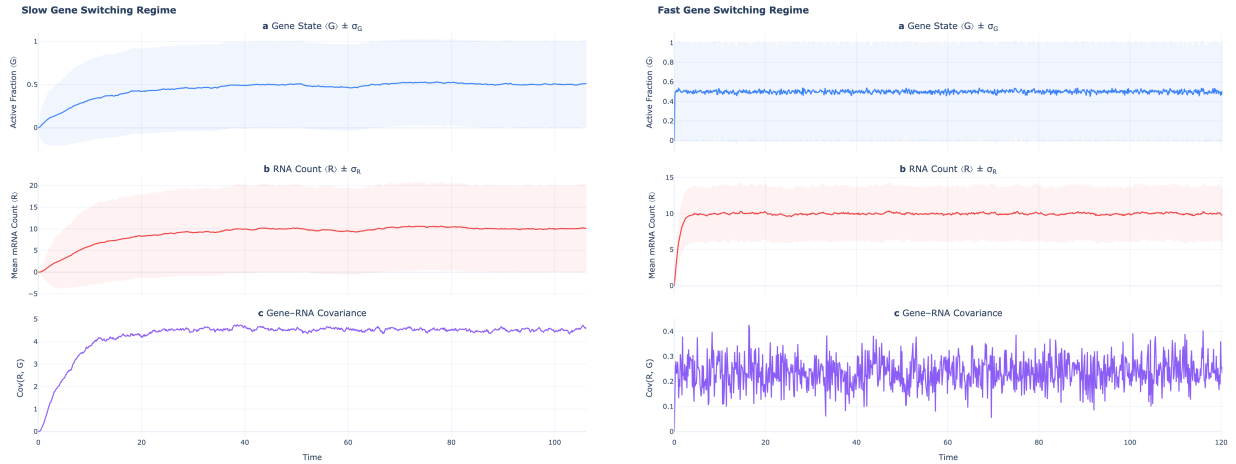


Figure 2: Switching speed reshapes noise at a fixed mean ($\mu_R \approx 10$). Slow switching (left, $k_{\text{on}} = k_{\text{off}} = 0.05$) produces long bursts, a wide $\pm\sigma_R$ band, and strong gene–RNA covariance. Fast switching (right, $k_{\text{on}} = k_{\text{off}} = 10$) flickers so quickly that the promoter averages out, leaving a narrow band and near-zero covariance.

Conclusion. The SSA reproduces every qualitative feature expected of the telegraph model, and perfectly matches the quantitative closed-form predictions for the Fano factor.

4.2 SSA vs. ODE: Moment Dynamics (Edge 1, Notebook 2)

Aim. Verify that the stochastic simulation and the deterministic moment ODEs describe the same dynamics at all times (Edge 1, Section 3.4). If they agree, the SSA ensemble averages should track the ODE curves up to sampling noise of order $1/\sqrt{N_{\text{rep}}}$.

Result. The direct test is the overlay in Figure 3. The SSA ensemble averages perfectly track the deterministic ODE curves throughout the transient relaxation and into the steady state, despite the two methods sharing no underlying code. The only visible deviations are minor finite-sample fluctuations in the SSA covariance curve, which shrink as N_{rep} increases. Crucially, the broad $\pm\sigma$ bands reflect genuine biological cell-to-cell variability—which the telegraph model is designed to capture—rather than numerical error, matching the analytical spread predicted by the moment ODEs.

This agreement is not limited to the symmetric case. As detailed in Notebook 2, the SSA robustly tracks the deterministic dynamics across all parameter regimes, including slow and fast switching, high and low expression, and varying initial conditions. Even in the hardest case—the high-noise, slow-switching regime (Figure 4)—the match remains exact.

While visual overlays confirm qualitative agreement, Section 3.4 requires the deviation to fall quantitatively as $1/\sqrt{N_{\text{rep}}}$. Figure 5 tests this rate directly by measuring the root-mean-square (RMS) deviation of $\widehat{\mu}_R(t)$ from the ODE curve over the entire time grid. We use the RMS rather than the supremum, as extreme-value statistics do not yield clean power laws even under exact convergence. Plotted against N_{rep} on log-log axes, the deviation falls exactly along a straight line. The fitted slope over three decades is -0.521 , closely matching the $-1/2$ predicted by the Law of Large Numbers.

Conclusion. The SSA converges to the exact moment dynamics at the precise rate predicted by the Law of Large Numbers. Edge 1 therefore holds across all tested regimes. Furthermore, the $\pm\sigma$ bands do *not* collapse onto the mean as N_{rep} grows: they correctly measure persistent cell-to-cell variability as predicted by the variance equation (8).

4.3 Steady-State Validation (Edges 2 & 3, Notebook 3)

Aim. Verify that the long-time limit of the moment ODEs agrees with the independent steady-state sampler (Edge 2), that the sampler’s error decays at the expected Central Limit Theorem rate ($1/\sqrt{N_{\text{rep}}}$), that the stationary *distribution* reached by the SSA matches the sampler (Edge 3), and that both reduce correctly to known limits at the boundaries of parameter space.

Result. The ODE solutions in Figure 3 plateau at $\mu_G^{ss} = 0.5$ and $\mu_R^{ss} = 10$, perfectly matching the analytical fixed points (20). The direct sampler reproduces these values to within standard Monte Carlo error at $N_{\text{rep}} = 2 \times 10^5$. Figure 6 makes this convergence quantitative: on log-log axes, the mean absolute deviation of the sampler’s μ_R^{ss} estimate follows a precise $-1/2$ slope over four decades, confirming the $1/\sqrt{N_{\text{rep}}}$ decay rate.

A more rigorous check requires the sampler to reproduce *all five* steady-state moments simultaneously. Table 2 compares the sampled moments against their closed-form expressions (derived

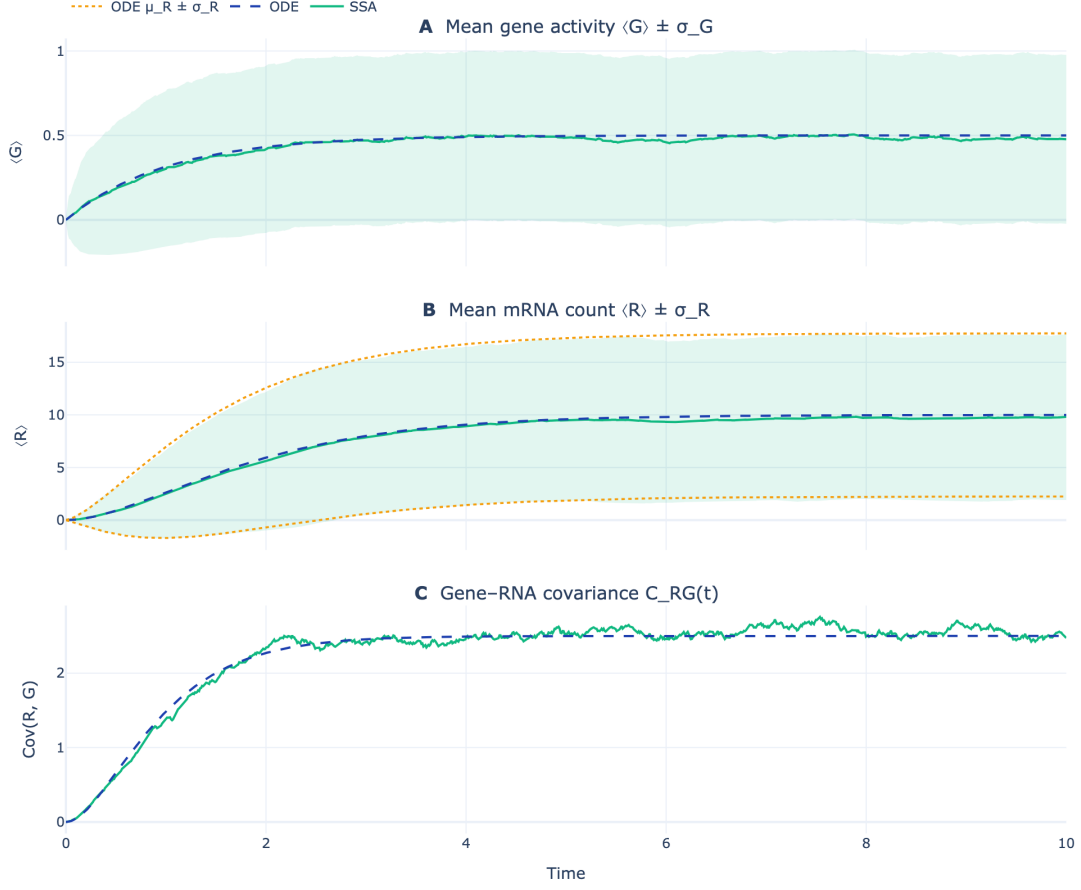


Figure 3: SSA ensemble averages (green, with empirical $\pm \sigma$ fluctuation bands shaded) and the deterministic ODE solutions (dashed blue) at the reference parameters from $(g_0, r_0) = (0, 0)$, with $N_{\text{rep}} = 1000$. In the mRNA panel, the analytical spread from the moment ODEs, $\mu_R \pm \sigma_R$, is drawn as amber dotted lines on top of the SSA band.

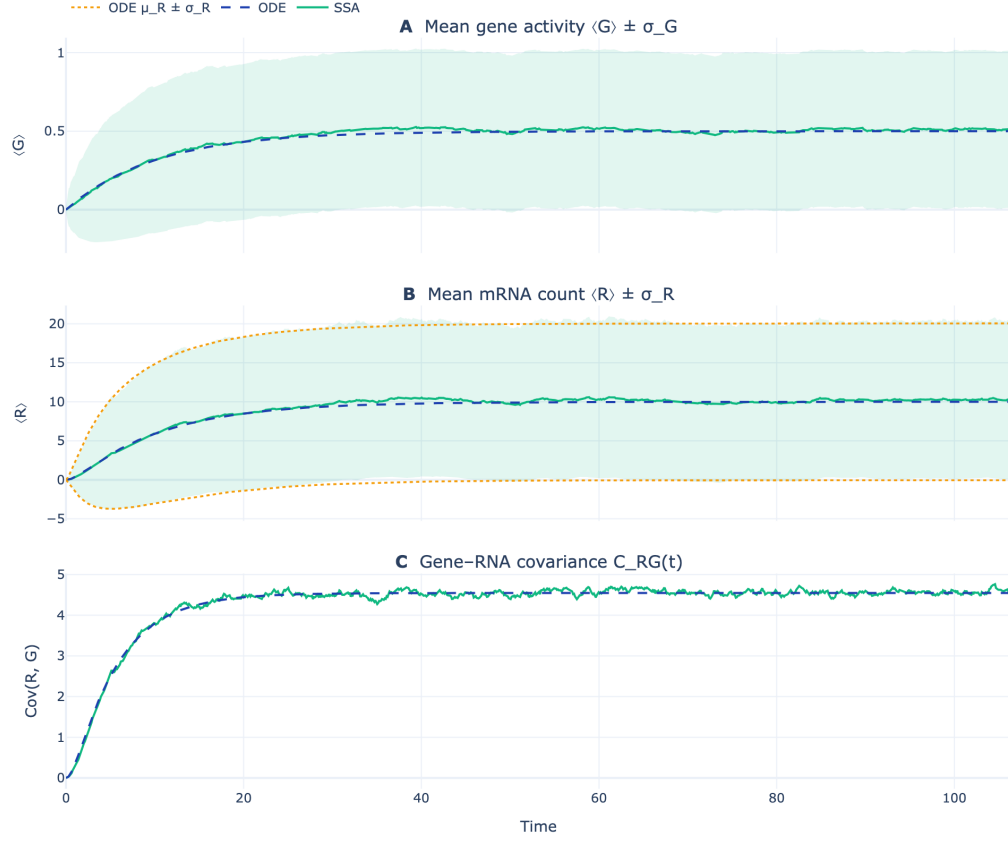


Figure 4: SSA versus moment ODEs in the high-noise slow-switching regime. Formatting follows Figure 3. Even where fluctuations are largest, the SSA ensemble averages track the deterministic ODE curves throughout the transient and into steady state, confirming Edge 1 beyond the symmetric reference case.

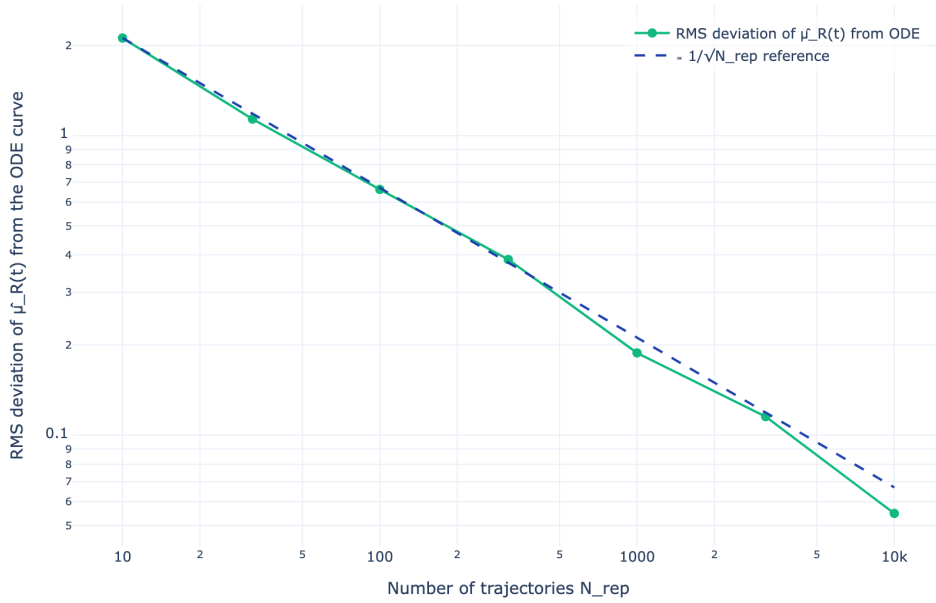


Figure 5: Convergence of the SSA ensemble mean to the moment-ODE solution (Edge 1) at the reference parameters. The statistic is the RMS deviation of $\widehat{\mu}_R(t)$ from the ODE curve over the full time grid; the dashed line is a $1/\sqrt{N_{\text{rep}}}$ reference anchored at the smallest ensemble. The fitted log-log slope is -0.521 , against the predicted $-1/2$. This is the Edge 1 counterpart of Figure 6, which tests Edge 2.

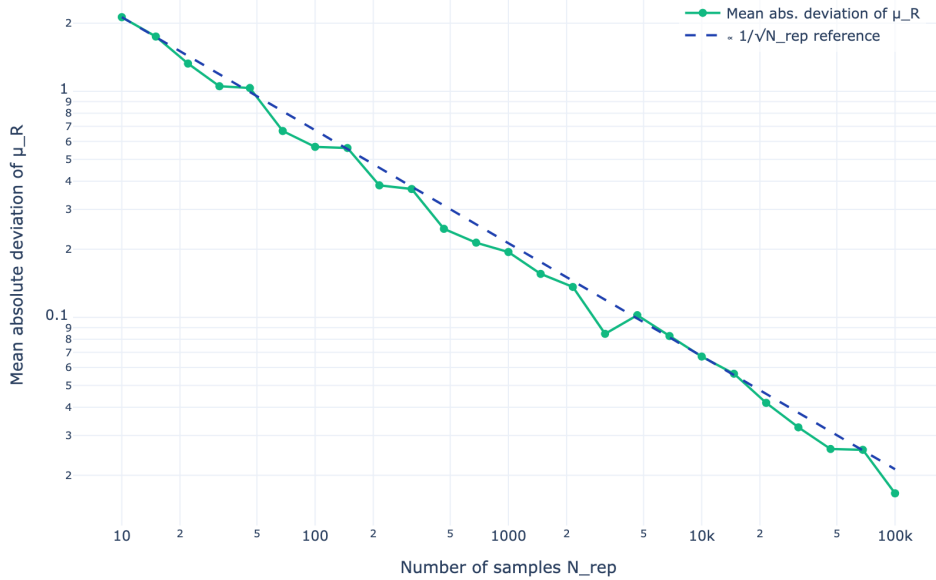


Figure 6: Convergence of the exact steady-state sampler's empirical moments to the analytical steady-state values as a function of the ensemble size N_{rep} . The error (mean absolute deviation) decays as $1/\sqrt{N_{\text{rep}}}$, in strict agreement with the Central Limit Theorem.

previously in (17) and (20)) for an asymmetric parameter regime. At $N_{\text{rep}} = 2 \times 10^5$, all absolute deviations are intrinsically small. When normalized by their respective statistical standard errors, the differences represent unremarkable sampling fluctuations rather than systematic bias. Furthermore, the absolute error in μ_R continuously follows the $1/\sqrt{N_{\text{rep}}}$ scaling down to 0.0145 at $N_{\text{rep}} = 10^5$.

Moment	Analytic	Sampler	Abs. diff.	Std. error
μ_G^{ss}	0.3333	0.3317	0.0016	0.0010
μ_R^{ss}	6.6667	6.6277	0.0389	0.0142
σ_G^{ss}	0.4714	0.4708	0.0006	0.0004
σ_R^{ss}	6.4979	6.4767	0.0212	0.0105
C_{RG}^{ss}	1.7778	1.7666	0.0111	0.0071

Table 2: Steady-state moments from the direct sampler ($N_{\text{rep}} = 2 \times 10^5$) against exact analytical values for $k_{\text{on}} = 0.5, k_{\text{off}} = 1, k_{\text{syn}} = 20, k_{\text{deg}} = 1$. The final column is the Monte Carlo standard error obtained via bootstrapping. Because all five moments are computed from a single array of Beta draws, their deviations are strongly correlated. For instance, the mean mixing fraction of this specific draw falls ≈ 0.0016 low; since $\mu_R = (k_{\text{syn}}/k_{\text{deg}})\bar{p}$ multiplies this shortfall by 20, it accounts for most of the observed deviation in μ_R^{ss} . Thus, one underlying fluctuation drives the differences across multiple rows.

Finally, we test the boundary limits where the model reduces to classical results. As predicted in Section 3.3, when the gene never switches off ($k_{\text{off}} = 0$), the telegraph model collapses to a simple birth–death process with a Poisson steady state [13, 14]. Here, the moment ODEs correctly relax to $\mu_G^{ss} = 1$ and $\mu_R^{ss} = k_{\text{syn}}/k_{\text{deg}}$. The sampled counts overlay the analytic Poisson PMF perfectly (Figure 8), yielding a Fano factor of ≈ 0.99 against the theoretical value of 1. In the opposite extreme, where degradation is disabled ($k_{\text{deg}} = 0$), mRNA accumulates indefinitely and no steady state exists; the sampler robustly detects this divergence and raises a `ValueError` rather than returning invalid data.

Edge 3 as a distributional test. Two different distributions can share the same mean, variance, and covariance. To rigorously claim that the three descriptions agree, we must compare the full stationary distributions directly, not merely their first two moments. Since both endpoints of Edge 3 produce samples, we compute the Total Variation (TV) distance between them.

Figure 7 compares these distributions in two regimes. For each, 20 000 independent SSA trajectories are run far past stationarity. To avoid autocorrelation and understating the spread, the mRNA count is recorded at a single late time point per replicate. This histogram is compared against 2×10^5 draws from the steady-state sampler. The discrepancy is measured using the TV distance, $\text{TV} = \frac{1}{2} \sum_r |\hat{P}^{\text{SSA}}(r) - \hat{P}^{\text{smp}}(r)|$. Unlike a χ^2 test, TV requires no arbitrary binning of near-empty states, is bounded in $[0, 1]$, and directly represents the maximum probability by which the two laws can differ on any event.

The null distribution for the TV statistic is generated via *direct resimulation*. Assuming the null hypothesis (that both methods sample the exact same stationary law), we simulate 200 fresh pairs of samples at the same sizes and recompute the TV. This directly evaluates the test’s theoretical claim without resampling approximations. The common alternative—bootstrapping each side from its own empirical measure—is intentionally avoided here: it adds sampling noise twice over and centers the null band above the true expected TV, resulting in an overly permissive test.

The slow-switching regime (Figure 7, right) provides the most demanding test, precisely where

moment-matching is least informative. The exact stationary law is strongly bimodal: mass concentrates at $R = 0$ (long OFF periods) and near $R = k_{\text{syn}}/k_{\text{deg}} = 20$ (long ON periods), while the mean $\mu_R = 10$ falls in the trough between them. A method could perfectly match μ_R and σ_R yet completely fail to capture this shape. The $\pm\sigma$ bands, while useful summaries elsewhere, falsely invite a Gaussian interpretation that this underlying law strictly rejects.

Measured via TV, the two distributions agree. In the slow regime, $\text{TV} = 0.0117$ against a null median of 0.0130 (95th percentile 0.0166, one-sided $p = 0.74$). In the symmetric regime, $\text{TV} = 0.0093$ against a median of 0.0155 (95th percentile 0.0191, $p = 1.00$). Neither regime exhibits a discrepancy beyond expected sampling chance.

The symmetric case yields a TV strictly below all 200 null draws ($p = 1.00$). Rather than indicating an anomaly, this simply reflects a highly central draw: its empirical variance of 59.8 matches the exact analytic $\sigma_R^2 = 60$ to within 0.4%, and splitting the sample in half reproduces the dispersion of two independent sampler draws. The agreement is fortunate in magnitude but expected in kind. Crucially, the null band is narrow enough that a TV of 0.03 would have triggered a clear rejection. This confirms the validity of the SSA—a hand-written nested loop and the most error-prone component of the project—against the verified, symbolic steady-state sampler.

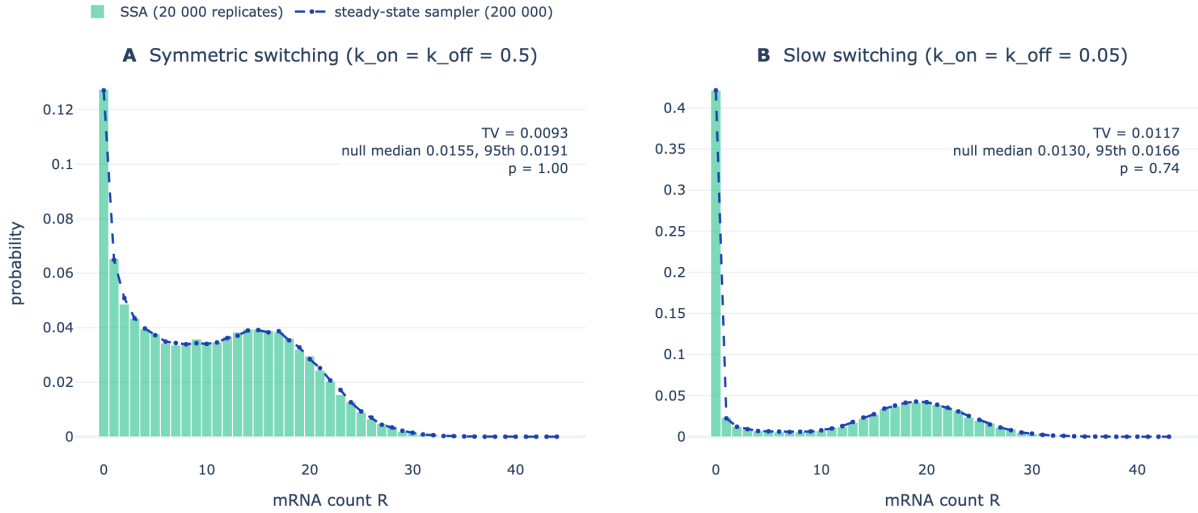


Figure 7: Edge 3 as a distributional test: the stationary mRNA distribution from 20 000 independent SSA trajectories (green bars, one sample per replicate) against 2×10^5 draws from the steady-state sampler (dashed blue). Left: symmetric reference regime. Right: slow switching ($k_{\text{on}} = k_{\text{off}} = 0.05$), where the distribution is strictly bimodal. Each panel reports the observed Total Variation (TV) distance, the median and 95th percentile of the resimulated null band, and the one-sided p -value.

Conclusion. The sampler is an unbiased, $\sqrt{N}$ -consistent estimator of the steady-state moments (Edge 2). Furthermore, Edge 3 holds independently rather than by inheritance: Figure 7 demonstrates that the exact agreement extends to the full stationary distribution in a strict test that had ample statistical power to fail. The accurate collapse to classical limits ($k_{\text{off}} = 0$ and $k_{\text{deg}} = 0$) confirms this agreement.

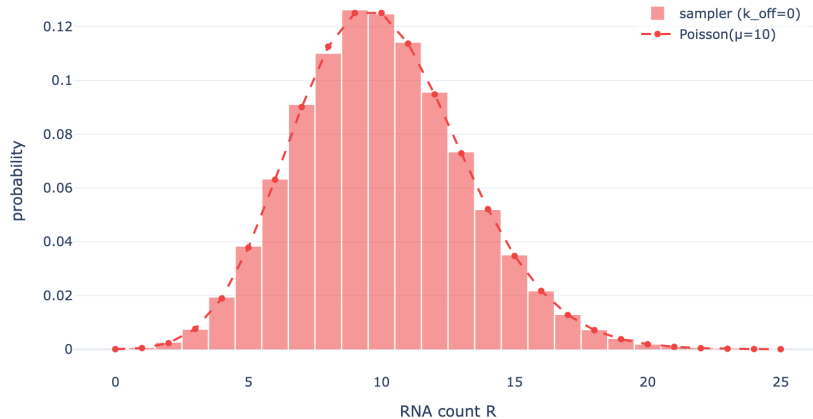


Figure 8: Edge case $k_{\text{off}} = 0$ (permanently active gene). As predicted theoretically, the sampled steady-state mRNA distribution precisely matches a Poisson PMF with mean $k_{\text{syn}}/k_{\text{deg}} = 10$ (Fano factor ≈ 1).

5 Conclusion

The steady-state sampler is an unbiased, $\sqrt{N}$ -consistent estimator of the steady-state moments, successfully satisfying Edge 2. Furthermore, Edge 3 holds independently: Figure 7 demonstrates that the exact agreement extends beyond the first two moments to the full stationary distribution under rigorous statistical testing. Together, these three edges close the validation triangle, establishing the strict mutual consistency of the stochastic, deterministic, and stationary descriptions. Finally, the exact collapse to classical limits ($k_{\text{off}} = 0$ and $k_{\text{deg}} = 0$) confirms that this structural agreement is highly robust and holds globally across the parameter space.

6 Future Work

While the telegraph model provides a rigorously validated baseline, two extensions would capture broader biological complexity.

Incorporating the protein level. The natural next step is a three-stage model (gene $\rightarrow$ mRNA $\rightarrow$ protein) that includes translation at rate k_{trans} and protein degradation at rate $k_{\text{deg},p}$. The analytical and simulation tools validated here carry over directly: the SSA naturally accommodates the two additional reactions, while the moment ODE system expands to capture protein variance and mRNA–protein covariance.

Gene regulatory networks. In vivo, gene products often toggle other genes, forming complex regulatory networks. Such architectures can be described by mapping the interacting genes to an *Ising model*, where ON/OFF states act as spins and regulatory links as couplings. The primary obstacle is the state-space explosion: for n genes, the joint distribution spans 2^n discrete promoter configurations (multiplying the unbounded mRNA and protein spaces), rendering direct integration intractable. Bridging this gap requires approximate high-dimensional density estimators, such as neural networks.

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